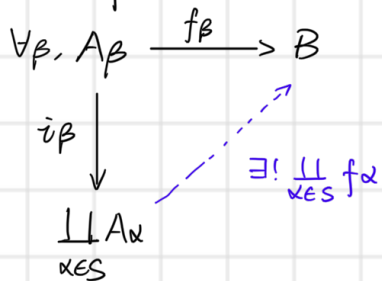


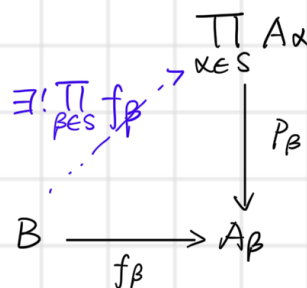
Product Topology

Universal properties:

① disjoint union



② product



Goal: Consider topological spaces and continuous maps.

Equip $\coprod A_\alpha$ and $\prod A_\alpha$ with topologies s.t. the universal properties hold.

Def. The disjoint union topology on $\coprod_{\alpha \in S} X_\alpha$ is

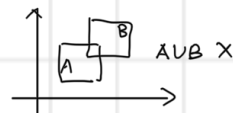
$$\coprod_{\alpha \in S} U_\alpha \text{ is open} \iff \forall \alpha \in S, U_\alpha \text{ is open in } X_\alpha.$$

Rmk. The universal property holds for $\coprod X_\alpha$ as spaces.

(ほかのはなし)

Consider $\prod_{\alpha \in S} X_\alpha$, X_α is a topological space.

① B' not topo-space:



$$\mathcal{B}' := \left\{ \prod_{\alpha \in S} U_\alpha \mid U_\alpha \text{ is open in } X_\alpha, \forall \alpha \in S \right\}.$$

① $\mathcal{B}'$ is not a topology

② The topology τ' generated by $\mathcal{B}'$ doesn't satisfy the universal properties.

ex. $f_\alpha: \mathbb{R} \rightarrow \mathbb{R} \times \mathbb{R} \times \mathbb{R} \times \dots, x \mapsto (x, x, x, \dots), \alpha \in \mathbb{N}.$

$$\prod f_\alpha: \mathbb{R} \rightarrow \mathbb{R}^{\mathbb{N}}, x \mapsto (x, x, x, \dots).$$

$\prod f_\alpha$ is NOT continuous under τ' , since

$$(\prod f_\alpha)^{-1}((-1, 1)^{\mathbb{N}}) = \{0\}. \text{ not open.}$$

Rmk. τ' is called the box topology on $\prod_{\alpha \in S} X_\alpha$.

Def. The product topology on the set $\prod_{\alpha \in S} X_\alpha$ is the topology generated by

$$\mathcal{B} = \left\{ \prod_{\alpha \in S} U_\alpha \mid U_\alpha \text{ is open in } X_\alpha, \forall \alpha \in S \text{ and } U_\alpha = X_\alpha \text{ for all but finitely many } \alpha \in S \right\}$$

Remark. $\tau_{\text{product}} \subseteq \tau_{\text{box}}$.
 $\nwarrow$ weaker / coarser.

Claim: The product topology on the set $\prod_{\alpha \in S} X_{\alpha}$ satisfies the universal properties for spaces.

(a) $p_{\beta}: \prod_{\alpha \in S} X_{\alpha} \rightarrow X_{\beta}$ is continuous.

(b) For any continuous maps $f_{\beta}: B \rightarrow X_{\beta}$, the map $\prod_{\alpha \in S} f_{\alpha}: B \rightarrow \prod_{\alpha \in S} X_{\alpha}$ is continuous.

pf: (a) $\forall$ open set $U_{\beta} \subseteq X_{\beta}$, $p_{\beta}^{-1}(U_{\beta}) = \prod_{\alpha \neq \beta} X_{\alpha} \times U_{\beta}$ open in τ_{prod} .

Prnk. The product topology is the weakest topology on $\prod_{\alpha \in S} X_{\alpha}$, s.t.

$p_{\beta}: \prod_{\alpha \in S} X_{\alpha} \rightarrow X_{\beta}$ is continuous, $\forall \beta$

(b) It suffices to prove $\forall U \subseteq B$, $(\prod f_{\alpha})^{-1}U$ open.

We have $(\prod f_{\alpha})^{-1}(\prod_{\alpha \in S} U_{\alpha}) = \bigcap_{\alpha \in S} f_{\alpha}^{-1}(U_{\alpha})$

since $U_{\alpha} = X_{\alpha}$ for all but finitely many, when $U_{\alpha} = X_{\alpha}$,

$f_{\alpha}^{-1}(X_{\alpha}) = B$, vanishing in intersection;

$\Rightarrow (\prod f_{\alpha})^{-1}(\prod_{\alpha \in S} U_{\alpha}) = \bigcap_{\alpha \in S} f_{\alpha}^{-1}(U_{\alpha})$ is open as finite intersection of open sets, $\square$

Ex. $S' = \bigcirc$, $S' \times S' = \text{torus } T^2$.

Completion of metric spaces (geometry).

Recall. Given a metric space (X, d) ,

a sequence $(x_n)_{n \in \mathbb{N}}$ is a Cauchy sequence if

$\forall \varepsilon > 0, \exists N, \forall \alpha, \beta > N, d(x_{\alpha}, x_{\beta}) < \varepsilon$.

Def. A metric space is complete if any Cauchy sequence has a limit.

Ex. $(\mathbb{R}, |\cdot|)$ is complete. $(\mathbb{Q}, |\cdot|)$ is not.

Def. A continuous map $f: (X, d_X) \rightarrow (Y, d_Y)$ is an isometric embedding if $d_Y(f(x_1), f(x_2)) = d_X(x_1, x_2)$, $\forall x_1, x_2 \in X$. (automatically injective).

Def. An isometric embedding is a completion if (Y, d_Y) is complete and $f(X)$ is dense in Y .

Thm. (1) Any metric space has a completion.

(2) Completion is unique up to isometry.

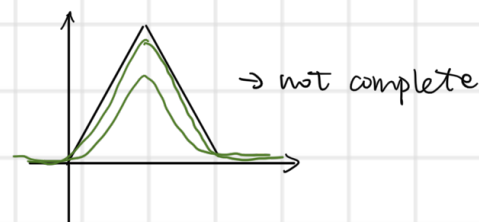
Ex 1. $(\mathbb{Q}, \|\cdot\|_p)$ p -adic norm.
 $\xrightarrow{\text{completion}} (\mathbb{Q}_p, \|\cdot\|_p)$

Ex 2. $C^\infty([0,1]) = \{ \text{smooth } f: [0,1] \rightarrow \mathbb{R} \}$.

(1) $\|f\|_{C^0} := \max_{x \in [0,1]} |f(x)|$

$(C^\infty([0,1]), \|\cdot\|_{C^0})$ is a metric space.

$\xrightarrow{\text{completion}} (C^0([0,1]), \|\cdot\|_{C^0})$



(2) We can also consider different norms on $C^\infty([0,1])$:

$\|f\|_{L^p} := \left(\int_0^1 |f(x)|^p dx \right)^{1/p}$

$(C^\infty, \|\cdot\|_{L^p}) \xrightarrow{\text{completion}} (L^p([0,1]), \|\cdot\|_{L^p})$

(3) $\|f\|_{W^{k,p}} := \left(\int_0^1 |f(x)|^p + |f'(x)|^p + \dots + |f^{(k)}(x)|^p dx \right)^{1/p}$

$(C^\infty, \|\cdot\|_{W^{k,p}}) \xrightarrow{\text{completion}} (W^{k,p}([0,1]), \|\cdot\|_{W^{k,p}})$

" $W^{k,p}$ - Sobolev"

Chapter 4. Compactness

Motivation: As sets S , any set maps $f: S \rightarrow \mathbb{R}$ has a maximum
 $\Leftrightarrow S = \text{a finite set.}$

Compactness is an analog of finiteness in topology.

Def. An **open cover** of a topology space X , is $\mathcal{U} = \{U_\alpha\}_{\alpha \in S}$ s.t.
 U_α is open in X , $\forall \alpha \in S$, and $\bigcup_{\alpha \in S} U_\alpha = X$.

Def. A topology space X is **compact** if every open cover of X has a finite subcover. A subset is **compact** if its subspace topology is compact.

Ex. (from $\mathbb{R}$) An subset $X \subseteq \mathbb{R}$ is compact $\Leftrightarrow X$ is closed and bounded.

Thm 1. If X is compact, then every continuous $f: X \rightarrow \mathbb{R}$ has a maximum.
 $\uparrow$

Lemma. If X is compact, and $f: X \rightarrow Y$ is continuous, then $f(X)$ is compact.

Thm 2. Suppose X is compact and Y is Hausdorff, then any bijective continuous map $f: X \rightarrow Y$ is a homeomorphism.

Lemma 1. X compact, $A \subseteq X$ closed, then A compact.

Lemma 2. Y Hausdorff, $B \subseteq Y$ compact, then B is closed in Y .

Thm 3. Suppose X is compact and first countable. Then any sequence $(x_n)_{n \in \mathbb{N}}$ in X has a converging subsequence.

Ex. (0) finite spaces is compact.

(1) closed & bounded subset in $\mathbb{R}^n$.

e.g. $S^n = \{x \in \mathbb{R}^{n+1} \mid |x| = 1\}$.

$D^n = \{x \in \mathbb{R}^n \mid |x| \leq 1\}$.

$T^n = \underbrace{S^1 \times \dots \times S^1}_n$ $\begin{matrix} x_0 & \text{---} \\ x_1 & \text{---} \\ x_2 & \text{---} \dots \text{---} \end{matrix} \cap x_n$
Cantor set

(2) $D^\infty := \{x \in \mathbb{R}^{\mathbb{N}} \mid \sum_{n=1}^{+\infty} |x_n|^2 \leq 1\}$. is not compact.

(3) Tychonoff theorem.

If X_α is compact $\forall \alpha \in S$, then $\prod_{\alpha \in S} X_\alpha, (\tau_{\text{prod}})$ is compact.

e.g. $[0,1]^{\mathbb{N}}$ is compact under τ_{prod} , but not under τ_{box} .

Thm. (Tychonoff theorem)

The product $X_1 \times X_2$ is compact $\Leftrightarrow X_1$ and X_2 are compact.

pf. $(\Rightarrow)$: $p_i: \underbrace{X_1 \times X_2}_{\text{compact}} \rightarrow \underbrace{X_i}_{\text{compact}}$ continuous

$(\Leftarrow)$: Let $\mathcal{U} = \{U_\alpha \mid \alpha \in S\}$ be a open cover of $X_1 \times X_2$.

claim: $\forall x \in X_1, \exists$ neighborhood V_x of x , s.t. $V_x \times X_2$ is covered by a finite subset of $\mathcal{U}$.

pf. $\forall y \in X_2, \exists \alpha(y), (x, y) \in U_{\alpha(y)} \in \mathcal{U}$.

By definition of $\tau_{\text{prod}}, \exists U_{1,y} \subseteq X_1, U_{2,y} \subseteq Y$ open

$$(x, y) \in U_{1,y} \times U_{2,y} \subseteq U_\alpha.$$

Note $\{U_{2,y} \mid y \in X_2\}$ is a open cover of X_2

It has a finite subcover since X_2 is compact.

$\exists \{U_{2,y_1}, \dots, U_{2,y_n}\}$. Let $V_x := \bigcap_{i=1}^n U_{1,y_i}$.